From One-Off Activity to Teaching Sequence

Seminar 3 • Playful AI with IoT and Generative AI

University of Copenhagen · Department of Computer Science

The problem with great workshop moments

You have all seen it — perhaps run it:

A brilliant activity. Students engaged, laughing, arguing.
Two weeks later: *"Wait, did we do something about AI?"*

Engagement is not retention. A moment is not a sequence.

This hour: what research says about the difference — while you are mid-design, when it can still change what you build.

Why one-offs fade: three findings

1. **Isolated episodes don't transfer.** A concept met once, in one context, stays welded to that context — the classifier is "the sticky-note game", not an idea about data.
2. **Novelty masks learning.** High engagement in a single session often reflects the *newness*, not the concept. Strip the novelty and test what remains.
3. **No return, no depth.** Understanding grows by *revisiting* an idea in a new form — not by meeting it once vividly.

The fix is not better one-offs. It is **sequences**.

What a sequence is (and is not)

Not: the same activity repeated.
Not: a topic block ("three weeks on AI").

A sequence is one core question, revisited through escalating encounters:

unplugged → plugged → applied → critiqued

Each encounter strips a simplification from the last — transposition, staged over time.

You already know one sequence: this workshop

Encounter	Form	What was removed
"Is this AI?"	argument	nothing — pure intuition
Sticky-note trees	paper model	vagueness: now there's a mechanism
Storytelling game	embodied model	determinism: now there's probability
Sensor + LLM bridge	real system	pretence: now it's the actual thing

The study-and-research angle

The Danish/French didactics tradition offers a sharper version:

Start not from content but from **a question worth investigating** –

"Can our classroom tell who is in it?"
"Why does the chatbot speak worse Danish than English?"
"What does the school building know about us?"

A good question **generates** the sequence: students need the concepts to pursue the answer. Need precedes explanation.

Anatomy of a sequence question

Test your driving question against three criteria:

1. **It requires the concept** — you cannot answer it without meeting classification, prediction, or data honestly
2. **It survives an answer** — the first answer opens two more questions
3. **It belongs to the students** — situated in their room, their language, their data

"What is machine learning?" fails all three.
"Can the micro:bit tell when our class is bored?" passes all three — and smuggles in ethics without announcing it.

Practical patterns for gymnasium reality

You do not control the timetable. Sequences that survive contact with a Danish gymnasium:

- **The 4 × 20-minute thread** — one encounter per week inside existing lessons; beats a single 90-minute event
- **The returning artefact** — the paper tree built in week 1 is *retested* in week 3 against a real model
- **The cross-subject relay** — the same question passes from biology (the data) to Danish (the language of the output) to samfundsfag (who is accountable)
- **The October move** — end the sequence by having students redesign the first activity for a younger class. Transposition as assessment.

What this means for your canvas — right now

After lunch, add one row to your design:

"Encounter 2: ___"

One sentence. Where does your activity go *next* —
what simplification does the second encounter remove?

You do not need a full sequence today.
You need a **direction** — proof your activity is a first encounter, not a dead end.

One line to keep

An activity is a good *first* encounter
when it leaves a question that
the *second* encounter is needed to answer.

Back to the canvas at 13:00.